

To Be Is to Make Oneself One: Finite Unity as Made, and How to Test It

Abstract

Metaphysics offers two accounts of what makes a finite being one. On the first, unity is given: a substrate persists and bears what changes. On the second, unity is derived: a stability within a flux that traverses it. The first presupposes what it means to explain; the second has no discriminant of individuated persistence. This paper argues that both fail on the case that should be easiest for the first, the stone. A finite determination is a difference held against a ground that, left to itself, does not sustain it; every exposed finite being drains, and nothing real escapes exposure. The stone therefore makes itself one, at a cost, and the pole of the given loses its paradigmatic case. What remains is a third position, which I state as an axiom: to be is to make oneself one. Unity is neither a state prior to the act nor a stability abstracted from it, but that act insofar as it holds. The axiom is testable. Perturb a finite being and locate where the irreversible trace of its response falls: on a margin of its own, and it is a *self*; on an external bearer's, and it is a *carriage*; on none, and it is an *aggregate*. Every finite being makes itself one; only a self remakes itself. I argue finally that for a being grounded on a finite margin this making is not optional, since inaction is not a stable state but dissolution.

Keywords: individuation; finite unity; substance; process; irreversibility; metaphysics of persistence

1 Two Accounts, and What Both Concede

1.1 The question

What makes a being one being? Not what there is, nor what kinds of thing the world contains, but under what conditions something holds as one individual rather than as a coordinated plurality, an effect of conceptual carving, or a stability of use. The question is classical without being exhausted, and its persistence is not accidental:

most answers presuppose what they mean to explain. To say that an entity is *this* entity, one needs a principle that individuates it; but if that principle is an essence, a form, or a sortal, it demands in turn to be individuated. To say that it persists through change, one must designate it as the same before and after, which presupposes the criterion of identity being sought.

Two families of answer occupy the field, and they are usually presented as rivals. I want to show that they concede the same thing.

1.2 Unity as given

On the first account, unity is prior to activity. Aristotle gives the position its lasting form: primary substance, what is neither said of a subject nor in a subject, is the first candidate for unity, and *this* Socrates exists independently of the qualities he bears (*Categories*; *Metaphysics* Z). The gesture prevents unity from dissolving into a multiplicity of predicates. But positing that there are primary substances does not say what makes a being hold as this subject. Substance names unity without exhibiting its operation.

Leibniz reasserts singularity in another idiom, at the cost of commitments that make individuation depend on postulates external to it: complete concepts, pre-established harmony, a theory of simplicity. The criterion is operative in the logical sense rather than the empirical one. It does not indicate how to test, on a concrete configuration, whether there is one. Strawson (1959) inherits the problem without the theological resources and reformulates it descriptively, asking how we identify and re-identify particulars within a shared spatio-temporal framework. The answer is precise but epistemically oriented; the ontological question is defused by translation rather than resolved. One can know how we identify an individual without knowing what makes it one.

Contemporary substance ontologies retain the essential commitment. Wiggins (2001) makes identity through change turn on the sortal concept under which a thing falls, so that what a thing is fixes the conditions under which it remains that thing. Lowe (1998, 2006) grounds persistence in the possession of dispositions that need not be exercised: what persists does so by what it *would* do, not by what it does. Koslicki (2018) revives a hylomorphic answer in which form is the principle by which a substance is one, which meets the presuppositional worry head on and answers it by taking form as an explanatorily prior constituent. Whatever else divides these positions, they agree that there is something the finite being is, beneath or before what it does.

1.3 Unity as derived

On the second account, unity is posterior to activity. The individual is a temporary crystallization within a becoming (Rescher 1996); organisms are processes actively maintained at several timescales (Nicholson and Dupré 2018); persistence is the persistence of processes rather than of things (Meincke and Dupré 2020). The gain is real. The inert substrate is abandoned, and change ceases to be a problem requiring a bearer.

The loss is a discriminant. If unity is a stability in a flux, then any sufficiently stable pattern qualifies, and the account owes an answer to the question of which stabilities individuate. A whirlpool maintains a form through a complete turnover of its matter; so does a cell. Under a criterion of processual persistence they are on a footing. Processualism can describe the difference between them but cannot derive it, because the resource that would derive it, something belonging to one and not the other, is what the account has given up.

I leave the pair here and return to it in §5. The test proposed there is directed at this pole as much as at the first, and the whirlpool is where it does its work: what separates it from the cell is neither a degree of stability nor of renewal, and the criterion has to say what it is.

1.4 The shared concession

The two positions look symmetrical, and their symmetry is the point. Both distribute being and doing across two planes; they differ only over which plane comes first. The substantialist places identity before activity: there is a thing, and then operations that conserve it. The processualist places activity before identity: there is a flux, and then stabilities within it. Neither considers that the distribution is itself the error.

This is not a merely formal observation, because the distribution is what produces both failures. If there is first a thing and then its operations, the thing must be individuated independently, and the presupposition returns. If there is first activity and then its stabilities, nothing in the activity marks one stability off from another, and the discriminant is missing. Remove the distribution and both failures lose their source.

1.5 Two senses of “one”

One distinction must be announced before the argument begins, because without it §2 will be misread. There are two senses of “one”, and they do not coincide. In a minimal sense, every finite being makes itself one: it holds as a determination, under

constraint and at a cost. In a strong sense, only some beings *remake* themselves one, regenerating their own conditions through a cycle of their own, whereas most made unities come undone as soon as they are perturbed. The stone makes itself one and does not remake itself; the organism does both.

The argument of §2 bears on the first sense, and establishes it universally. The test of §5 bears on the second, and discriminates. Nothing in §2 is therefore a claim that the stone and the organism hold as one in the same way. The question of what separates them is deferred, not levelled.

A note on method. The continental tradition has reached a version of the first diagnosis. Simondon (2005) argues that any ontology setting out from the constituted individual to seek its principle presupposes what it means to explain, and that individuation must be thought from a pre-individual field. The diagnosis is just, and I take it as independent confirmation. But it remains a diagnosis: transductive individuation names the structure classical ontologies lack without furnishing a procedure that decides a concrete case. My principal interlocutor is therefore the analytic literature, for its discipline of explicit criteria and limit-cases.

2 The Stone

2.1 Why this case

The argument of this section is directed at the first pole, and it is conducted on the terrain that pole would choose. The case against inert givenness is easy to make with organisms, and worthless for that reason: no one holds that a bacterium is a substrate on which properties are imprinted. The substantialist's strongest instance is the apparently inert, closed, propertied thing, the stone or the rock that has lain in the same place for ten thousand years and appears to require nothing of itself in order to remain what it is. If the stone turns out to be a made unity, the pole of the given does not lose a peripheral example. It loses its paradigm.

2.2 A boundary is a boundary only maintained

Begin with what it is to be finite. A finite determination is a difference held against a ground that, left to itself, does not sustain it. To be finite is to be bounded, to be *this* rather than the surrounding indistinction. But a bound does not posit itself; a boundary is a boundary only insofar as it is maintained, and the surroundings have no reason to respect a line they do not draw. An unsustained difference is therefore perpetually to be renewed, or it fades.

This much is conceptual, and true a priori of anything finite. It is legible on finitude alone and requires no physics. Call it the *task*: the standing requirement that a bounded difference in grip on a ground be renewed in order to remain that difference.

2.3 Two magnitudes under one word

The task is not yet a cost, and the two must be separated, because the substantialist's best reply exploits their conflation.

Under the word "cost" hide two magnitudes. The first is modal and static: what it would take to undo the determination, which for a crystal is the depth of its potential well. This magnitude is a priori in the sense just given, being the task read as a quantity. The second is dynamic: maintaining the determination *expends* something each cycle, so that a drain is levied and not recovered. This second magnitude is not read a priori. A perfectly reversible world would know the same task without the least drain, since differences would still have to be held and holding them would cost nothing.

That holding drains follows from our world being irreversible, of which the second law is one expression. This is a posit about the world rather than a conceptual necessity, and I mark it as such: it is judged downstream, by what it lets one see, not on its intrinsic evidence.

2.4 The crystal, and why it does not save the given

The distinction just drawn gives the substantialist his sharpest objection, and I state it at full strength. Take a perfect crystal at absolute zero. The static cost remains, since it would still take work to undo it, but the drain is nil. At equilibrium, nothing further is expended. Here the two magnitudes divorce, and here the substantialist appears to recover his inert given: a finite being that is, without having to do anything at all.

He does not recover it, for two reasons, and the second is the one that matters.

The first is that the crystal does not exist. Absolute zero is inaccessible; every real crystal is at finite temperature, hence agitated, hence draining by diffusion, creep, and sublimation, however infinitesimally. And abstracting from the thermal, no structure escapes, at cosmological timescales, the background radiation traversing it or the slow decay of its matter. Where the thermal drain would vanish, another nameable mechanism restores it. Zero drain is the never-attained limit of a series of beings that all drain.

The second reason is that the ideal crystal is not merely unrealized but incoherent as described. To be finite and to be exposed are one property rather than two. A finite is a difference bounded against a ground, hence in grip on that ground, hence

exposed to it. A finite with no grip on any ground would be bounded against nothing, which is to say it would be the Whole, or a pure possible, and not a finite. The ideal crystal is constructed by removing exposure while retaining finitude, and that operation cannot be carried out. It is not a counterexample to the claim but a non-instantiated idealization obtained by deleting a constituent of the very concept it instantiates.

The anti-substantialist claim is thereby made precise rather than lost. It is not that everything is dynamic in a sense the ideal crystal would falsify. It is that every exposed finite being drains, and that nothing real escapes exposure.

2.5 What is established, and what is not

Established: no real finite being is inertly given. The stone is not a block of being laid down once and for all, but a determination in grip on a ground that wears it, and therefore indexed to time and to a milieu rather than indifferent to both. Its unity is made. The pole of the given has lost the case it would have chosen.

An equivocation must be refused here, because the word “cost” carries two magnitudes that the argument must not be allowed to trade on. For a being with a cycle, expenditure buys continuation: the reserves drawn down purchase the next round. For the stone, nothing of the sort occurs. Its bonds sit in a potential well and are not renewed by any work, and the drain it undergoes is not a price paid for holding but the erosion of what holds. A substantialist is entitled to press exactly this: the depth of the well is granted, and the dynamic drain measures the stone’s failure to hold rather than its holding.

I grant the point and narrow the claim accordingly. What §2 requires is not that the stone pays for its unity, but that no real finite being is exempt from exposure. The inert given demands a determination indifferent to the ground it is bounded against, and there is none: exposure is not an accident of the stone but a constituent of finitude. That is what “makes itself one” means in the minimal sense of §1.5, and it means no more than that. Whether anything buys its own continuation is a further question, which §5 answers and §2 does not.

Not established, and the debt should be exact. That the non-sustaining is *directed*, that an unrenewed finite declines rather than reinforces itself, is taken here as a structural trait of the irreversible world and not derived from anything more primitive. Its ultimate reduction to a combinatorial or thermodynamic condition, and the treatment of recurrence over long timescales, exceed this paper. The argument of §2 therefore has two grades: the task is conceptual, the drain is posited, and the conclusion inherits the weaker of the two.

What §2 establishes belongs to the minimal sense of §1.5. That the stone does not *remake* itself is a further fact, which §5 turns into a test.

The two poles are now both retired: the given has lost its paradigm here, and the derived was shown in §1.3 to lack a discriminant. What remains is not a compromise between them but the position they jointly exclude.

3 The Position That Remains

3.1 A dichotomy, not a survey

Both available accounts distribute being and doing across two planes and differ only over which comes first. The one that places identity first has now lost the case it would itself have chosen; the one that places activity first cannot say which continuities of activity are the activity of one being.

What follows is not that some third position has been proved. It is narrower and more useful. The disjunction *either being and doing are distributed across two planes, or they are not* is exhaustive, and both orderings of the distribution have now been shown to fail, each for a reason internal to it. Whatever survives must therefore deny the distribution itself. That is a constraint on the space of remaining positions, not yet an occupant of it.

The axiom is the occupant I propose:

To be is to make oneself one.

For every finite being, unity is not a state prior to the act but the act itself insofar as it holds. Not, on one side, a thing that is one, and on the other the operations that maintain it: one plane, an act whose unity is the continuous enactment of its own holding.

I want to be exact about what the preceding sections license. They license the denial of the distribution. They do not license this particular way of denying it, which is a substantive proposal and could be wrong while the diagnosis stands. What the axiom must now do is earn the difference between itself and a mere negation. It must have content, and that content must make a difference to cases. §4 states the content; §5 puts it to a test that it can fail.

3.2 Why each term is load-bearing

Every word in the formulation is doing work, and the quickest way to show it is to remove them one at a time. *To be* names finite actuality, not existence in the

unrestricted logical sense; the axiom is a claim about finite determinations, and I take no position on whether anything else falls under it. *To make* marks constitution rather than fabrication, and nothing in it implies an agent who first exists and then produces itself, a reading treated in §3.3. *Oneself* marks that the act and what is constituted in it are not two independently individuated terms. *One* marks that the act is not sheer occurrence but holds as a determination.

Now remove them. “To be is to act” describes a world of activity in which nothing individuates, and the question of §1 returns unchanged: why call some regions of this activity beings? “To be is to persist” reinstates a bearer given before what it does, and §2 has removed its best case. “To be is to make oneself” is generalized self-production, processualism with a reflexive verb, and it loses the constraint that would separate the cell from the storm.

The operative word is therefore *one*. What it does not mean must be said. It does not mean numerical simplicity: an organism has trillions of components and is one in the relevant sense. It does not mean indivisibility, nor impermeability, nor the absence of nested cycles. It means that the act holds as a determination, so that changes can be attributed to the same finite unity rather than merely redescribed as successive states of an unrestricted field.

This is why I speak of unity-in-act. Unity is neither deposited beneath the activity nor extracted from it afterwards. The act is one insofar as it holds, and the being is that holding insofar as it occurs.

3.3 “Oneself” without a prior self

An objection threatens the axiom before it is deployed, and it bears on two words. The reflexive “oneself” appears to posit a subject that makes and an object on which the making falls, hence a subject prior to the act, which is what the axiom refuses. And “one” redoubles the threat: to make one, is this not for a unifier to gather a unified, reintroducing beneath unity the duality of what unifies and what is unified? Under this double reading the axiom reinstates twice the division it means to lift.

The objection reads “oneself” as reflexive. The construction I mean is middle, in the technical sense of comparative grammar. Benveniste (1966) characterizes the middle voice as designating processes in which the subject is itself engaged in what it accomplishes rather than being the external agent of an action falling on a distinct patient: *gignomai*, to be born, *phainomai*, to appear. The subject is the site where the process takes place, not a term standing outside it.

Read in that voice, both threats dissolve. The subject is not given before the act, since it is in the act as its site; and “one” does not presuppose a unifier distinct from the

unified, since unification is the mode under which the act holds and not an operation performed on a previously multiple matter. To make oneself one, in the middle voice, is to hold as one without the holding and the held coming apart.

Grammar does not settle ontology, and the middle voice does not show that unity so conceived is real rather than verbal. Its role is narrower and sufficient: the sentence need not carry the circularity a reflexive reading imposes on it, so the objection does not dispose of the axiom before the argument begins. The burden falls on the test of §5, which the axiom can fail.

One restriction, which will matter later. At the level of the axiom, “oneself” names no interiority, no agency, no intention, and no self in any strong sense; it marks only the non-externality of a finite being to its own enactment. That every finite being makes itself one is what the axiom says. That some finite beings are selves is not, and requires the further structure of remaking.

3.4 How an axiom is to be judged

Calling this an axiom is not a way of exempting it from argument, and I should be explicit about the standard I am accepting. An axiom is an explicit primitive, and the question to put to it is not whether it is self-evident, since it is not, and neither is any of its rivals. The question is whether it discriminates cases its rivals leave undecided, at a cost in commitments one can state. I have stated the commitments in §3.2 and will add the rest in §4. The discrimination is §5’s business, and I have built that section so that it can come out the wrong way.

What follows immediately is the content. The axiom compresses three claims that it mobilizes jointly and that are not derivable from one another. Taken separately, each falls back into a position §1 has already removed.

4 What the Axiom Commits One To

The axiom compresses three claims. They are mobilized jointly, they are not derivable from one another, and their solidarity is what makes the displacement. I state them here because §5 depends on all three, and because a reader is entitled to see the bill before the test.

4.1 Constitutive activity

Nothing finite is inertly given. Every finite being is bounded against a ground that wears it, and to cease absolutely to make itself is to cease to be. This is not the claim that everything transforms visibly, repairs itself actively, or buys its continuation with

work; it is the claim, narrowed in §2.5, that no finite determination is indifferent to what it is bounded against. What it excludes is the finite being as an inert substrate on which properties are imprinted. It is the weakest of the three, and the one for which §2 supplied an independent argument.

4.2 The identity of being and doing

The second is stronger, and it does not follow from the first. That a finite determination presupposes constitutive activity does not yet forbid holding that there is first a thing and then operations that conserve it, a bearer that acts rather than a bearer given inertly. The axiom refuses this residual distribution as well.

Doing is not an attribute added to being. It is its modality. Correlatively, being is not a silent support beneath doing; it is that doing insofar as it holds as one. This is the content that separates the axiom from substantialism, which gives the substrate as primary, and from radical processualism, which gives the flux as primary without stabilizing what in it would be. The middle voice (§3.3) is the grammatical resource that makes the identification statable without circularity; it is not what makes it true.

This content is the one §5 leans on hardest, and the reason belongs here. If being and doing were distinct, a thing on one side and its maintenance on the other, then the question of whose margin bears the cost would be a question about where an accountant draws a line. One could redescribe the whirlpool as a being whose maintenance operations happen to be performed by the river, and the organism as one whose operations happen to be performed internally, with no ontological difference between them, only a difference in the boundary chosen for the audit. The criterion would then record a convention.

Under the identity, that redescription is unavailable. The whirlpool is not a thing that has its holding paid for elsewhere. It *is* the holding, and if the holding is the river's doing, there is no second being there that borrows. The difference between bearing one's own cost and having it borne is therefore not a fact about where an external boundary is drawn, but about whether the act traces a boundary at all. This is what will make the verdicts of §5 more than a bookkeeping choice, and it is the point at which the axiom stops being an ornament on the criterion and becomes its condition.

4.3 The monism of the act

If the finite being is never other than the doing by which it holds as one, there are not two orders of reality, an inert ground and a dynamic regime, a matter and a mind, to be articulated afterwards. There is a single category of which all finite beings are

modalizations.

This is not a reductionism. It does not say that all beings resemble one another, but that their differences must be thought as differences of regime within one type of being rather than as differences between two irreducible orders. Without it, the trichotomy of §5 would be three analogies rather than one criterion: the stone, the organism, and the whirlpool would be sorted by three unrelated standards, and the claim that a single question discriminates them would have no ground. With it, the same question is well-posed across the range, and the answers differ because the regimes differ, not because the beings belong to different worlds.

Monism is also what keeps the axiom from being a biological criterion in disguise. If self-maintenance marked a separate order of being, “to be is to make oneself one” would be a thesis about living things with an unexplained remainder for everything else. It is instead a thesis about finite determination as such, of which self-maintenance is one regime.

4.4 Why the three hold together

The solidarity can be checked by removing each in turn. Activity without the identity of being and doing makes activity a superadded attribute, and a substrate reappears beneath it. Identity without monism produces a dualism in which being-doing holds of one category and something else holds of the rest, and the criterion loses its range. Monism without activity produces indistinct static entities, and individuation collapses into the question of how to carve one homogeneous stuff. Together the three draw one configuration: a finite being exclusively, non-dually, and costingly in act, without a reserve of inertia and without a division of planes.

One consequence must be recorded, because it is what the test will exploit. If every finite being makes itself one, no finite being is exempt from the question of where the cost of its holding falls. The question is universal in application and, this is the substantive part, not universally answerable in the same way: the drain is real for every exposed finite being, but a cycle that takes the drain in charge is not. Whether anything answers to “who pays” is what differs across cases, and it is the differing that §5 reads.

5 The Test

5.1 What renewal does not see

Take the two cases left standing in §1.3. An organism and a whirlpool both present a stable form traversed by a complete turnover of their matter. Both persist through

the replacement of their components; both would be described, in the vocabulary of processual persistence, as organizations actively maintained across time. Under a criterion of renewal alone they are indiscernible, and a theory bounded to such a criterion must either class them together or decide between them by appeal to something it has not stated.¹

The form of this objection needs care. It is not that processual accounts are unable to notice the difference. They notice it, and describe it well. It is that the resource which would *derive* the difference is the one such accounts have given up. If unity is a stability within a flux, then what individuates a stability can only be a feature of the stability: its degree, its duration, its complexity, its causal integration. None of these separates the cases. The whirlpool is stable, persists indefinitely under a maintained flow, has an intricate internal structure, and is causally integrated in the strong sense that every part of it depends on every other. What it lacks is not a further degree of anything on that list.

Rescher (1996) posits the individual as a temporary crystallization within becoming; Nicholson and Dupré (2018) treat organisms as processes actively maintained at several timescales; Meincke and Dupré (2020) locate persistence in processes rather than things. Each is a defensible account of how identity survives change, and none of them, on its own resources, determines which of two equally maintained organizations is the one being. That is not a failure of care but a consequence of having placed activity before unity, which is the second horn of the distribution §1.4 identified.

5.2 The test

What is not visible under renewal is visible under perturbation.

By a perturbation I mean any event that displaces a finite being from its nominal regime and forces it to mobilize whatever capacity it has to respond: a thermal shock for a cell, a material failure for an artifact, a supply breakdown for an organization. What makes such an event a test is not its physical nature but its structure. The response it demands leaves a trace that is not restored when the perturbation ceases. The cell that has synthesized its heat-shock proteins does not recover the energy spent. The wound closes as a scar and not as the prior tissue. Returning from B to A is not the erasure of the passage from A to B, but a further transformation with a cost of its

¹The pair is borrowed from a debate it does not belong to me to settle here. The question of what makes a biological individual one has its own literature, in which anatomical, genetic, and evolutionary answers have each been pressed and found wanting on ordinary cases (Godfrey-Smith 2009; Clarke 2010; Pradeu 2012; Bouchard and Huneman 2013). My use of the case is metaphysical rather than biological: the organism and the whirlpool are the cleanest instance of two organizations equally maintained and unequally individuated, and nothing in the argument turns on the organism being alive.

own.

The cost at issue is therefore the dynamic drain of §2.3, not the static magnitude that measures what it would take to undo a structure. This matters, because only the first is levied somewhere in particular. What it would cost to shatter a crystal is a fact about the crystal; what it costs to respond to a perturbation is an expenditure drawn from a determinate resource, and the criterion asks which one.

The question is then precise and empirically addressable: *on what resource, belonging to what, is the irreversible trace levied?*

One restriction, and it is a restriction rather than a promise. What follows is a decision procedure whose inputs are empirical: it says what to look for, and the world can answer it against me. It is not a measurement protocol. I supply no quantity, no unit, no threshold, and no case in which the measurement has been carried out, and nothing in the argument turns on my doing so. Specifying an observable that satisfies the criterion in a given system is a separate task, and a harder one than stating the criterion. What the test does presuppose is the drain posited in §2.3: if that posit fails there is no trace to locate, and this section fails with it. Beyond that it assumes nothing about which beings have margins, which is what it is built to find out.

5.3 Three answers, and two privations

Three answers to that question exhaust the field, and the exhaustiveness is structural rather than enumerative (Figure 1). Either some cycle takes the trace in charge, or none does. If none does, the trace disperses and accumulates. If some cycle does, either it draws on a resource of its own or it draws on another's. Where the cost falls on a margin proper to the being, the being is a *self*. Where it is borne by an external infrastructure or bearer, the being is a *carriage*. Where no cycle takes it up at all, the being is an *aggregate*.

Carriage and aggregate are not two further positive kinds standing alongside the first. They are two privations. The carriage has the cycle and lacks the endorsement; the aggregate lacks the cycle, hence lacks any *who* that could pay. This asymmetry disposes of a circularity that would otherwise threaten the question itself. "Who pays?" has a determinate answer only where a cycle traces one. For a carriage the answer is traced by an observer who decides where to draw the audit; for an aggregate the question is without object, since an aggregate is not a unity for which one must decide who pays. To ask who pays, of an aggregate, is already to presuppose in it the self it lacks.

The grain of the trichotomy is coarse, deliberately. Within carriage, the distinction between a carried form and the bearer that carries it admits a refinement that

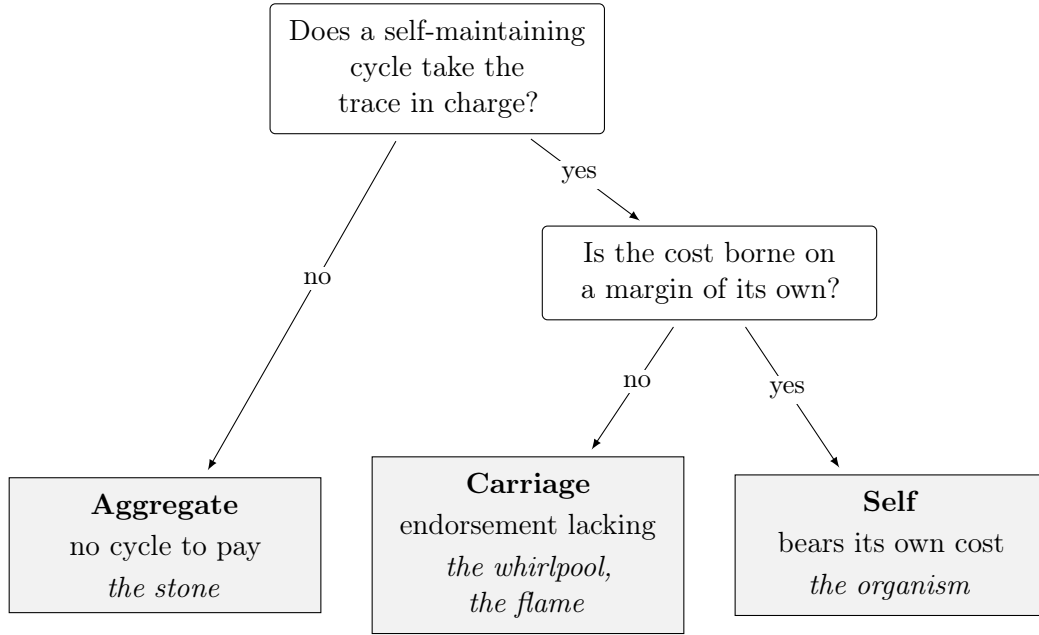


Figure 1: The criterion as a decision procedure. Two questions and two privations: the aggregate lacks the cycle, the carriage lacks the endorsement. The first branch is read at a scale on which dissolution is slow. The second question is settled empirically, by locating where the irreversible trace of a perturbation falls.

produces further verdicts.² I set it aside here, since the three regimes suffice for the discrimination §1.3 demanded, and the refinement would import commitments the present argument does not need.

5.4 Verdicts

The four verdicts are collected in Table 1.

The organism. A bacterium under thermal stress synthesizes heat-shock proteins, restructures its membrane, and reallocates its metabolic budget. The irreversibility of that response falls on its own margin. Reserves are drawn down, the response is paid for from within, and what remains afterwards is a system that has been altered by having responded. The scar is the general form of this, the material evidence that someone paid. Verdict: self. Its unity is not postulated; it is paid on a margin that belongs to it.

²The refinement matters because carriage as defined here covers two unlike situations: a cost borne by a passive infrastructure, as the river bears the whirlpool, and a cost borne by another self, as a host bears what lives on it. The mitochondrion and the lichen are the cases that force the distinction, and they are also where stratified endorsement makes “who pays?” admit answers at more than one level at once. Nothing in the present argument turns on which way they are decided.

	Cycle	Where the trace falls	Verdict
The stone	none	on the material itself, where it accumulates	aggregate
The organism	regenerative: produces the conditions of its next cycle	on its own metabolic margin; the scar is the record	self
The whirlpool	non-constitutive: maintains a form it does not produce	on the fluid traversing it, and away downstream	carriage
The flame	consuming: metabolizes without repairing	on the pre-given fuel	carriage

Table 1: The four cases under the two readings. The organism and the whirlpool are indiscernible under renewal alone and separate on the third column. The flame is the case in which the response to perturbation resembles an aggregate’s while the cost profile is a carriage’s, and in which the cost profile arbitrates.

The whirlpool. The reconfiguration falls on the fluid traversing it. Perturb the flow and the form adjusts, but the adjustment is carried away downstream; remove the flow and the whirlpool disappears without having endorsed anything. In the physics of Prigogine and Stengers (1984), the whirlpool is a dissipative structure, an order maintained far from equilibrium by a flux traversing it. I take up that description entirely and add the verdict thermodynamics alone does not render: such orders are real, and they are carried. The river pays. Verdict: carriage, a unity that is genuine and borrowed.

The stone. No cycle of its own restores its bonds. Perturbed, the trace accumulates: the chip stays, the crack propagates, and nothing takes the cost in charge. Verdict: aggregate. Here the equivocation announced in §1.5 must be held off, since to classify the stone as an aggregate is not to say it is not one. “Aggregate” names the absence of a self-sustaining cycle, not the absence of unity, and dissolution is its default regime. The criterion does not discriminate one against not-one, which the axiom settles universally, but, among beings that make themselves one, those that remake themselves on their own margin.

The flame. One case takes a paragraph of its own, because it separates a criterion from a slogan. The flame has a real feedback loop: heat melts the wax, vaporizes the fuel, and sustains the flame that melts the wax. Its causal chain closes. But it closes thermally rather than constitutively. The flame consumes the wax without ever producing it; its cycle metabolizes without repairing. Under perturbation, remove the fuel, and it does not compensate. It burns what remains until extinction, as an

aggregate would. Verdict: carriage, with the pre-given fuel as payer. What the flame shows, and what §5.6 will use, is that having a loop and having a loop that produces the conditions of its own operation are two different things, and that only the second bears on individuation.

5.5 Composition, and the life answer

The verdicts bear on a question analytic metaphysics has posed in another form. Under what conditions do several things compose one thing? Answers range from the universalism that lets any plurality compose (Sider 2001) to restrictions of various strengths, with conservative defences of the ordinary furniture in between (Korman 2015). The framing treats unity as a relation between a plurality and an object.

Van Inwagen (1990) gives the answer closest to the present one, and the proximity is instructive. Composition occurs, on his account, when the activity of the simples constitutes a life. This makes an activity rather than a relation the condition of composite unity, which is the same displacement I am proposing. The divergence is over what makes an activity constitutive. In van Inwagen's account, life is where the answer stops: asked why a life composes and a maintained vortex does not, one can point to the biological and no further. The endorsement criterion says what the appeal to life was standing in for, namely that the activity bears on its own margin the cost of its own resumption, and it separates the organism from the whirlpool by a property neither of them wears as a kind term.

The gain is that the criterion is structural rather than sortal. Two distant grains of sand compose nothing, for want of a common endorsement, even where an arbitrary mereological fusion unites them; and an entity may fall outside the biological and still endorse, or fall inside it and not. A full engagement with restrictivism, universalism, and priority monism requires a separate treatment. What I retain is that “who pays?” discriminates where “which pluralities compose?” remains indifferent.

5.6 Why “its own margin” is not circular

The obvious objection is that the criterion smuggles in what it was meant to produce: to say that the resource *belongs to* the being is to have given oneself the being in advance, since the boundary that decides whose margin it is, is the boundary the criterion was supposed to trace.

The objection bites if ownership is read as possession, and read that way the criterion fails at once, since the organism too depends on what it does not make. It dissipates entropy into its milieu, imports amino acids and cofactors it does not synthesize, and would die within minutes of isolation. Autarky cannot be the criterion,

because dependence on a milieu is universal and discriminates nothing.

What is required is a property of the cycle itself, statable without a possessor and without a prior boundary: the margin is proper when the effects of a reconfiguration re-enter, within the being, into the conditions of the next cycle. The heat-shock protein the cell has synthesized modifies the state that conditions its subsequent metabolism, and the loop closes inside. In the whirlpool, the rearrangement of the fluid exits into the river and conditions no cycle of its own; in the flame, the consumed wax conditions the next moment of burning but is not among the things the flame makes. Which effects re-enter which conditions is a mereological property of the cause-effect chain, requiring no pilot, no possessor, and no boundary drawn in advance. The circularity falls there.

I borrow the property rather than construct it. It has been developed under the names of operational and organizational closure by Maturana and Varela (1980), Varela (1979), Rosen (1991), Montévil and Mossio (2015), and Moreno and Mossio (2015).³ I take it here only in the role the objection requires, as what makes “its own” statable non-circularly, and not as a rival criterion of individuation.

This is also where §4.2 is discharged. Under the identity of being and doing, bearing one’s own cost rather than having it borne is not a fact about where an observer draws the audit. The whirlpool is one only for an observer who traces the boundary of who pays; the organism traces that boundary itself, since its being *is* that holding and the holding is what closes. Strong individuation is the existence of an answer to “who pays?” traced by the structure rather than by the observer, which is what an endorsed closed cycle furnishes and what both privations lack.

5.7 What organization does not settle

The property borrowed in §5.6 belongs to a family of accounts that read the organization of a system, and the family is wider than closure. Bickhard (1993, 2009) distinguishes systems that are *self-maintenant* from those that are *recursively self-maintenant*. A candle flame is his canonical case of the first: it holds itself above combustion threshold, vaporizes wax, and induces the convection that brings in oxygen, and so contributes to the conditions of its own persistence. What it cannot do is maintain the property of being self-maintenant. Deprived of wax it has no alternative to select, because it can only burn. Campbell’s bacterium is the canonical case of the second: it swims while heading up a sugar gradient and tumbles when it finds itself

³Whether closure so understood is sufficient for endorsement, that is, whether a cycle can produce all the conditions of its own operation and still have its costs borne elsewhere, is a further question, with consequences for the individuality debate in the philosophy of biology. It does not bear on the circularity objection, for which closure is borrowed here, and I take it up in §5.7.

heading down, differentiating the two conditions and selecting accordingly.

The proximity is close enough that the difference has to be stated, and stating it is informative, because the two partitions cross rather than coincide. For Bickhard the flame occupies a genuine positive category; here it is a privation. The whirlpool, which contributes nothing to the conditions it depends on, falls outside his lowest positive grade, whereas here it sits alongside the flame. And a system that switched among self-maintenance strategies while drawing every unit of the cost from outside would sit at the top of his hierarchy and receive the verdict carriage here. The two criteria read different data: his reads what a system can do, and mine reads what it pays. Neither is a redescription of the other.

That last case is what shows the cost axis to be doing work of its own rather than tracking organization at one remove. Consider a chemical network that produces every catalyst its own reactions require, held in a continuous-flow reactor: substrates enter, products leave, the internal cycle is complete, and the network may well switch between alternative routes as the feed composition changes. On Rosen's criterion it is closed; on Bickhard's it is recursively self-maintenant. Perturb it, and it compensates on nothing of its own. The excursion is absorbed by the feed, and restoring the feed returns the network to its regime having levied no margin. Verdict: carriage.

Two claims must be separated here, and I separate them. Whether any such system is realized in nature is an open empirical question, and I do not assert that one is. That organization alone, however richly characterized, cannot deliver the verdict is not an open question: closure reads the catalyst graph, recursive self-maintenance reads the repertoire of selectable strategies, and neither says on which margin a perturbation is paid for. Enriching the organizational description adds constraints; it never adds a margin. This is why the cost is not a tracer of organization, and why §5.6 borrowed closure for the narrow job of making "its own" non-circular rather than adopting it as the criterion.

5.8 Paid unity

The thesis of the first axis can now be stated in its final form.

A finite being holds as one in the strong sense if and only if it endorses, on a margin of its own, the cost of regenerating its own conditions.

Three components, articulated. *Endorsement*: the cost is levied and located, not stipulated. *Closure*: the cycle's effects re-enter its own conditions, which is what makes "its own" non-circular. *Regeneration*: the cycle does not merely hold the being in a state but reconstitutes the conditions under which it can continue.⁴

⁴The closest kinship, and the sharpest demarcation, is with Deacon (2011). His teleodynamics

This is not a criterion posited alongside the axiom. It is what “to make oneself one” promised in §3, rendered testable by the empirical localization of irreversibility, and it is what separates the two senses of “one” announced in §1.5.

It remains to say how the criterion could come out the wrong way, since a test that cannot fail is not a test. Two conditions would defeat it. The first is that the irreversible trace of a perturbation should fall in the same place for the organism and for the whirlpool, so that, examined carefully, the cell’s response turned out to be paid for by its medium in the way the whirlpool’s is paid for by the river. The criterion would then discriminate nothing, and the axiom would have gained no explanatory advantage over the accounts of §1. The second and more serious is that the location of the trace should vary with the observer’s carving, so that by choosing a different system boundary one could make the same physical episode come out as endorsed or as borne. Endorsement would then be a convention after all, and §4.2 would have failed to secure what it claimed. A technically armed version of this objection is available, and I do not dispute its premise: in stochastic thermodynamics the apportionment of entropy production between a system and its surroundings is relative to the coarse-graining chosen. The criterion does not rest on that apportionment. It asks whether the irreversible trace of a response re-enters the conditions of the being’s next cycle, which is a question about which effects condition which and not about how a budget is divided. That is why I hold that it does not vary with the observer’s carving; but the claim is empirical in form, and it is the place where the account is most exposed.

6 Is Any of This Necessary?

One could grant everything in §5 and still hold that it describes a contingency. Beings in our world happen to have margins; some cycles happen to be arranged so that their effects re-enter their own conditions; the criterion sorts what it finds. On this reading the axiom would be a summary of local furniture, and “to be is to make oneself one” would be a fact about how things have turned out rather than a claim about finite being. This section argues that the reading is unavailable, and it argues it in a way that can be checked, by constructing the being for which the necessity would not hold and showing what it costs to be that being.

rests on the same pairing of work against a slope with the self-maintenance of a constraint against its degradation, and his passage from morphodynamics to teleodynamics runs close to the passage from carriage to self. But Deacon grounds the teleodynamic orientation on a constitutive absence, which has earned him the objection that an absence cannot do causal work. The present account does not carry that burden, since its principle is a positive cost levied on a finite margin rather than a lack.

6.1 The reductio

Suppose the identity of being and doing were contingent, so that it were conceivable for a finite being to be without doing, or for a doing to take place without being. There are two cases, and each ruins the axiom for a reason already established.

A being without doing reinstates the inert substrate. There would be finite being inertly given, which §4.1 excludes and §2 argued against on the case least favorable to it. Every finite determination holds under constraint, is to be renewed, and to cease making itself is to cease to be.

A doing without being introduces a flux with no bearer. There would be two orders of reality, pure activity on one side and being on the other, which §4.3 excludes, since there is a single category of which all finite beings are modalizations.

The identity of being and doing is therefore not an exchangeable stipulation. Its negation is not false by observation; it is incoherent with the contents the axiom mobilizes. But this establishes less than it appears to, and §6.2 says why.

6.2 The separating model

The reductio invites a deflationary reply, and the reply is a good one. If “to cease acting is to cease to be” follows from the way the terms have been set up, then the necessity is verbal: one has defined a being as an act and then discovered that an act which stops has stopped. Nothing about the world has been learned.

The reply is answered by constructing the alternative, and the alternative is coherent.

Consider two acts.⁵ The first is self-grounded. It has no external ground, its capacity to act is endogenous, and the margin on which its holding is paid is its own. For this act, not paying the cost is exhausting itself, and inaction is not a stable option but dissolution. The second is contingent in a precise sense: it has an external ground that compensates its cost. What becomes optional for it is not renewal but the *paying* of renewal, since an external source discharges the drain in its place, so that waste does not threaten its existence. It still makes itself, like every finite being. It simply does not pay what it costs.

The difference between the two is structural rather than verbal, and the second is constructible without contradiction. What renders the act necessary is therefore not

⁵I use “ground” throughout for the source from which the cost of holding is paid, and not for the relation of metaphysical grounding in the sense of Schaffer (2009) and Fine (2012). Nothing in the argument turns on whether the two coincide. In particular, the standard irreflexivity of the grounding relation is not infringed by a being that pays its own drain, since paying is not explaining. Whether self-grounding in the operative sense used here could be given an account in the technical one is a question I leave open.

logic alone but self-grounding. A being with an external ground escapes the necessity; a being grounded on a finite margin does not. Immanent necessity does not hold of everything conceivable. It holds of what grounds itself on a finite margin, which is the regime of the finite.

Note where this places the burden. The discriminating necessity, that one must pay on one's own margin on pain of dissolution where the contingent survives because another pays, runs on cost. It stands on the same side of the joint as the drain of §2.3, and is judged downstream in the same way.

6.3 Neither logical, nor external, nor contingent

The position can now be located among the terms modal metaphysics has fixed.

It is not logical necessity. "Making oneself one" is not a tautology, and its negation is not a formal contradiction but an incoherence with the axiom's contents. It is not causal or external necessity, since no transcendent instance imposes it and no external law grounds it. It is immanent, holding to the structure of what it characterizes.⁶

The two magnitudes of §2.3 reappear here as two necessities, and conflating them would overstate the result. The necessity of the *task* does not pass through cost: even in a perfectly reversible world the necessity of making oneself would remain entire, and only its price would vanish. That strand is a priori and thin, being near-analytic given the axiom. The necessity with discriminating content is the one §6.2 exhibits: it runs on cost and is posited rather than derived.

I state the result at the strength it has. Immanent necessity is thin on both sides of the joint, analytic upstream and posited downstream, and its value is not a modal discovery. Its value is that it blocks the deflationary reading with which this section opened. Endorsement is not a contingent regularity that our world happens to exhibit, but what it is for something to hold on a finite margin without an external ground.

6.4 The demarcation from Spinoza

The most likely misreading is that this is Spinozist necessitarianism in another idiom, and the demarcation is sharp enough to be made quickly.

The Spinozist *causa sui* is a self-grounding without cost. Substance is by its essence alone, and has nothing to maintain. The finite being grounds itself in an entirely different regime, operatively, on a margin it did not posit and that it exhausts, at a

⁶The position belongs to the anti-Humean family that anchors modality in objects rather than in a space of worlds (Bird 2007; Vetter 2015), and it offers an engine that family leaves open: the necessity of the finite act is not a primitive potentiality but has a structure, self-grounding on a finite margin under a cost that cannot be reduced to zero. I do not adjudicate among the members of that family here.

cost it must endorse without cease. It is self-grounded in the sense that no external instance pays in its place, which §6.2 makes precise. It is not *causa sui* in the sense that its existence would follow from its essence without its having to hold itself. Its necessity is not that of a self-caused existence but that of the act. The finite must make itself in order to be, and is never dispensed from paying for its holding.

The right Spinozist interlocutor is not substance in any case, but the finite mode. The *conatus*, the effort by which each thing perseveres in its being (*Ethics* III, 6–7), is already a self-maintenance under constraint, and it is the closest kin to the self as conceived here. The divergence is one of status. Spinoza posits the *conatus* as essence; the present account derives it from cost and finite margin. Allied on the self-grounding, distinct on the mechanism.

7 Scope

7.1 What is established

The argument has been deliberately narrow: an axiom, a criterion, and a relation between them. The criterion is that a finite being holds as one in the strong sense if and only if it bears, on a margin of its own, the cost of regenerating its own conditions, which is testable by locating where the irreversible trace of a perturbation falls and non-circular because “its own” is a property of the causal graph rather than of possession. Of the three, the relation is what I would defend first. Without the identity of being and doing, endorsement records where an observer draws an audit; with it, the difference between paying and being paid for is a difference in whether the act traces a boundary at all. That is what §4.2 secured and §5.6 discharged, and it is why the criterion is not detachable from the axiom that produced it.

7.2 Three grades of verdict

The criterion does not decide uniformly, and this is a property of the criterion rather than an embarrassment to be managed.

Some verdicts are determinate. The stone is an aggregate, the organism a self, the whirlpool a carriage. The endorsement structure is unambiguous, and the verdict is stable under reformulation.

Some are conditional: a dune renewed by the wind, a colony whose cycle may close at a level above its members. Here the criterion does not deliver a category but a decision procedure, and the verdict turns on parameters that can be identified and measured.

Some cases are indeterminate, and the criterion records the indeterminacy as a

result. Where endorsement is stratified or partly externalized, the conditions of application are not jointly satisfied, and there is no fact of the matter that the criterion is failing to detect.

Rival accounts decide uniformly, whether by a biological criterion, a mereological one, or a derivation from the whole, and thereby treat structurally ambiguous cases as though the ambiguity were a defect of the theory. I take it to be a property of the phenomenon, and prefer a criterion that says where it decides, where it conditions, and where it suspends.

7.3 What is not established

Four debts, stated rather than deferred.

The drain is posited rather than derived (§2.3). That an unrenewed finite declines rather than reinforces itself is taken as a structural trait of the irreversible world; its reduction to a combinatorial or thermodynamic condition, and the treatment of recurrence over long timescales, exceed this paper. The conclusions that depend on it inherit its grade.

That finitude entails exposure (§2.4) is a posit about the connectedness of the world, not a theorem. A plurality of causally disjoint finite islands is not obviously incoherent, and I have argued only that an island bounded against nothing would not be a finite; I have not shown that no other arrangement is available. The step is load-bearing for §2 and it is asserted.

The observer-independence of the trace's location (§5.8) is empirical in form. I have argued that what re-enters what is not a matter of choice, and this is where the account is most exposed.

The necessity of §6 is thin, and I have said so where it is stated rather than in a concluding qualification.

7.4 Final reformulation

The unity of a finite being is neither the instantiation of a substance nor the crystallization of a process. It is the regime in which an act holds as one by remaking itself, under the constraint of having to endorse its own cost. At this level, and only at this level, does the question “what is a being?” stop opposing what remains to what becomes, and become a more exact question, addressable in cases: what, in the finite, holds enough to pay for itself the price of its own holding?

Supplementary material

Several claims of this paper are machine-checked in Lean 4, and a separate supplement documents them. The independence of §5.7 is proved on record: closure and endorsement are encoded on disjoint data, all four combinations are inhabited, and a network that produces every catalyst it requires while carrying no margin receives the verdict carriage, so that the verdict is a function of the energetic profile alone whatever the catalyst graph. The remainder covers the dissolution-or-recurrence dichotomy behind the first branch of §5.3, together with the counterexample that bounds its scope; the determination of the regime by the cost profile alone (§5.8); the joint constructibility of the self-grounded and the externally grounded act, so that the necessity of §6.2 is not obtained by definition; and the drain of §2.3, which appears as a posited field rather than a derived proposition, corroborating the debt declared in §7.3.

The formalization is a consistency apparatus and not a proof engine. It establishes that stated structures are consistent and that stated conclusions follow from stated definitions; it does not establish that the encoded structures are faithful to the notions they are named after, which is defended by the argument and not by the compiler. The argument of §2 is not mechanized and is not a candidate for mechanization.

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